

NICOLÒ PAVIATO

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Faculty of Mathematics and Computer Science, Weizmann Institute of Science, ◊ Rehovot, Israel

I am a postdoctoral researcher in mathematics at the Weizmann Institute of Science, working with Prof. Omri Sarig. Prior to this, I was a postdoctoral researcher at the Hebrew University of Jerusalem with Prof. Mike Hochman. I received my PhD in 2023 from the University of Warwick, under the supervision of Prof. Ian Melbourne.

My research focuses on the statistical properties of deterministic systems, with a particular emphasis on the central limit theorem and its functional counterpart. More recently, I have also explored the typical behavior of random sequences of independently drawn digits in connection with Poisson genericity.



EDUCATION

University of Warwick, UK Ph.D. in Mathematics & Statistics	2019 - 2022
University of Warwick, UK M.Sc. in Mathematics & Statistics	2018 - 2019
University of Pavia, Italy M.Sc. in Mathematics	2015 - 2018
University of Pavia, Italy B.Sc. in Mathematics	2011- 2015

PUBLICATIONS AND PREPRINTS

Decay in norm of transfer operators for semiflows with I. Melbourne and D. Terhesiu - <i>Studia Math.</i> – arxiv.org/abs/2102.00026	2022
Nonexistence of spectral gaps in Hölder spaces for continuous time dynamical systems” with I. Melbourne and D. Terhesiu – <i>Israel. J. Math</i> – arxiv.org/abs/2104.04608	2022
A threshold for Poisson behavior of non-stationary product measures. with M. Hochman – <i>Ergodic Theory Dyn. Syst.</i> – arxiv.org/abs/2501.11423	2026
Rates for maps and flows in a deterministic multidimensional weak invariance principle – <i>Commun. Math. Phys.</i> – arxiv.org/abs/2406.06123	2026
Limiting behaviour of pattern counts in biased binary strings. with J.V. Kogan – <i>Israel. J. Math (to appear)</i> – arxiv.org/abs/2509.24654	2026

TEACHING

Teaching Assistant

- Ergodic Theory - 4th year maths *Warwick, 2022*
- Mathematical Analysis - 1st year non-maths " , 2021
- Stochastic Analysis - 4th year maths " , 2020
- Stochastic Processes - 2nd year statistics " , 2020
- Mathematical Analysis 3 - 2nd year non-maths " , 2019-2020
- Mathematics and Statistics - 1st year pharmacology *Pavia, 2016*

Supervisions

- I supervised two small groups of 2nd year maths students for ten weeks. *Warwick, 2019*

SELECTED TALKS

- Ergodic Theory and Dynamical Systems Seminars** (University of Bristol) *September 2026*
"Central Limit Theorem for Finitary Factors of iid" - *to happen*
- Seminario de Sistemas Dinámicos** (Pontificia Universidad Católica de Chile) *January 2026*
"Thresholds for Poisson Limits in Symbolic Dynamics"
- Action Now** (Technion - Israel Institute of Technology) *November 2024*
"A threshold for Poisson behavior of non-stationary product measures"
- Dynamical systems seminar** (Hebrew University of Jerusalem) *December 2023*
"Rates for a multidimensional weak invariance principle"
- 2nd Jangada Dinâmica** (Universidade Federal do Ceará, Fortaleza) *November 2022*
"Rates of convergence for a deterministic Weak Invariance Principle"
- Junior Ergodic Theory Meeting** (ICMS, Edinburgh) *March 2022*
"Absence of a spectral gap for flows"
- Ergodic theory and dynamical system seminar** (University of Warwick) *October 2021*
"Rates of convergence for a functional CLT"
- iPOD seminar** (Leiden University) *October 2021*
"Rates of convergence for a functional CLT"
- Cambridge joint CDT conference** (University of Cambridge, online) *December 2020*
"A family of continuous-time dynamical systems with a trivial CLT"

CONFERENCES/SCHOOLS

- Statistics, hyperbolicity and related problems** *March 2026*
(University of Warwick, Coventry)
- Beyond Uniform Hyperbolicity** *June 2025*
(ICTP, Trieste)
- Methods In Nonlinear Dynamical Systems and Ergodic Theory** *December 2024*
(University of Warwick, Coventry)
- Workshop on statistical properties of chaotic dynamics in and out of equilibrium** *August 2024*
(Erdős Center, Budapest)

School on probabilistic aspects of hyperbolic dynamical systems (Erdős Center, Budapest)	<i>August 2024</i>
Dynamical Systems: Geometric and Statistical Properties (University of Warwick, Coventry)	<i>July 2024</i>
Zurich Dynamic Conference (University of Zurich, Zurich)	<i>June 2023</i>
Zurich Dynamic School (University of Zurich, Zurich)	<i>June 2023</i>
Probabilistic methods in dynamics (Centro di Ricerca Matematica Ennio De Giorgi, Pisa)	<i>May 2023</i>
The Lorentz Gas (Lorentz Center, Leiden)	<i>December 2022</i>
2nd Jangada Dinâmica (Universidade Federal do Ceará, Fortaleza)	<i>November 2022</i>
International Conference on Dynamical Systems (IMPA, Rio de Janeiro)	<i>October 2022</i>
Conference celebrating the 100th anniversary of Rényi's birth (Hungarian academy of sciences, Budapest)	<i>June 2022</i>
Limit Theorems for Slowly Mixing Systems (ICMS, Edinburgh)	<i>June 2022</i>
Junior Ergodic Theory Meeting (ICMS, Edinburgh)	<i>March 2022</i>
Expanding dynamics: the school (Centro de modelamiento matematico, University of Chile, online)	<i>June 2021</i>
Spring school on transfer operators (classical and modern techniques) (Bernoulli center, Lausanne, online)	<i>March 2021</i>
Statistical aspects of geodesic flows in nonpositive curvature (University of Warwick, Coventry)	<i>January 2020</i>
One day ergodic theory meeting (Queen Mary University, London)	<i>November 2019</i>
Thermodynamic formalism: ergodic theory and geometry (University of Warwick, Coventry)	<i>July 2019</i>
Summer course in mathematics (Scuola matematica interuniversitaria, Cortona, Italy)	<i>July 2018</i>

VISITS

Leiden University - 1 week Host: Prof. Dalia Terhesiu	<i>October 2021</i>
Pontificia Universidad Católica de Chile - 2 weeks Host: Prof. Raimundo Briceño	<i>January 2026</i>
University of São Paulo - 1 week Host: Prof. Ian Melbourne	<i>February 2026</i>
University of Bristol - 1 week Host: Prof. Zemer Kosloff	<i>March 2026</i>

OTHER

Outside from Mathematics, I'm a sociable person who enjoys spending times with friends and appreciating my surroundings. I value diversity in both people and ideas, and I always enjoy a good conversation (especially over a glass of wine or beer). My hobbies include running, playing music, and staying active at the gym. In my downtime, I dedicate myself to learning Hebrew.